

ISMUN 4.0

BACKGROUND GUIDE



UNITED NATIONS SPACE

RESEARCH

ASSOCIATION

Agenda: *Mitigating the Proliferation of Orbital Debris, Securing Uncontrolled Satellites, and Preventing the Weaponization of Space Infrastructure.*

LETTER FROM THE EXECUTIVE BOARD

Dear delegates,

It is our great pleasure to welcome you to The United Nations Committee on the Space Research Association, here at the prestigious Intra School Model United Nations 4.0, 2026. It is our honour and privilege to serve as your Executive Board for the duration of the conference. This guide has been curated to serve as a starting point for research and will help you understand the basic concepts and issues revolving around the agenda.

Space Research is a key topic in this rapidly advancing world where multiplanetary projects are already ready to come in action. Debating on such an issue demands for your strong perspectives, solutions and absolute knowledge of the issues that follows the agenda. Your participation in the UNSRA will not only provide you with valuable insights into the workings of the United Nations but also offer you a platform to engage in meaningful debates and negotiations on a very interesting and pressing topic.

As members of the UNSRA Executive Board, we are here to support you throughout the conference. We encourage you to prepare thoroughly for the conference by researching your assigned country's positions, familiarizing yourself with the rules of procedure, and honing your public speaking and negotiation skills. Lastly, we would like to emphasize that the Model United Nations experience goes beyond the conference itself. It offers an opportunity for personal growth, cross-cultural understanding, and the development of lifelong skills that will serve you well in any future endeavour. This committee will be defined by your insights and contributions. We are excited to see the ideas you bring, the debates you drive, and the impact you leave behind.

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INTRODUCTION TO THE COMMITTEE

The United Nations Space Research Association (UNSRA) is envisioned as the body responsible for the operational safety and security of outer space. Its mandate centres on the practical, day-to-day realities of an increasingly crowded and contested orbital environment. Where much of international space law has historically been concerned with broad questions of sovereignty and the legal status of celestial bodies, UNSRA, is meant to turn these general principles into clear technical standards, monitoring systems, and specific rules about unacceptable actions.

The overall mandate of UNSRA aims at strengthening the operational and legal regime governing safety and security in outer space, resulting in improved conditions for tracking, accounting for, and safely managing objects placed in orbit. The Committee is further tasked with coordinating these efforts across national space agencies, private operators, and international bodies, recognizing that no single actor can secure the orbital environment alone. In doing so, the committee aims to build a more predictable and enforceable system of orbital conduct, reducing the ad hoc, case-by-case responses that have characterized space governance to date.

The Committee also draws on technical frameworks that, while not legally binding, shape accepted practice in orbit, including the Inter-Agency Space Debris Coordination Committee (IADC) guidelines and the UN Space Debris Mitigation Guidelines (2007).

INTRODUCTION TO THE AGENDA

Outer space may seem far away, but it plays a role in almost everything we do. Satellites guide ships and aircraft, carry phone and internet signals, forecast the weather, support disaster relief, and help national governments with security. Most people never notice this, yet modern life depends heavily on a small number of objects orbiting the Earth. This dependence is now at risk. Orbit is becoming crowded with debris, many satellites are no longer under anyone's control, and space technology is increasingly being used for military purposes. These three issues are closely linked, and this committee has been called to address them together.

1. Orbital Debris

I. *What It Is?*

Orbital debris refers to any human-made object in space that no longer serves a purpose. This includes dead satellites, old rocket parts, and fragments left over from past collisions or explosions.

- Debris travels at very high speed, so even a small piece can seriously damage or destroy a working satellite.
- A single collision can break one object into many pieces, each becoming a new hazard.
- More collisions mean more debris, which increases the risk of further collisions.

II. *The Kessler Syndrome*

Scientists warn of a situation called the *Kessler Syndrome*, where debris keeps causing more debris through repeated collisions. It can be compared to a road accident that is never cleared away — every new crash adds more chaos, until the road becomes unusable. If this continues in our orbit, some regions of space would eventually become too dangerous to use safely.

2. Uncontrolled Satellites

I. *The Problem*

An uncontrolled satellite is one that has run out of fuel, stopped working, lost contact with its operators, or has been abandoned. Unlike active satellites, it cannot move out of the way of danger, so it becomes a long-term risk to other satellites nearby.

II. *Why Is It Hard to Fix?*

- Under international law, a satellite remains the property of the country that launched it, even if it is broken or abandoned.
- No other country is allowed to move, capture, or interfere with it without permission.
- Attempts to remove or repair another country's satellite could be seen as a hostile act, even if the intention is safety.

This creates a difficult balance between keeping space safe and respecting each country's rights over its own property.

3. Weaponization of Space Infrastructure

I. *Peaceful Use, Militarization, and Weaponization*

Space is not only used for science and communication; it is also important for national security. It is very important to understand these three terms when talking about space weaponization:

- **Peaceful use** – space activity meant for civilian or scientific purposes.
- **Militarization** – using satellites to support military operations, such as communication or surveillance.
- **Weaponization** – developing or using weapons that can damage or destroy objects in space.

II. *Anti-Satellite (ASAT) Weapons*

ASAT weapons are designed to disable or destroy satellites. Some physically destroy their target, creating large amounts of debris. Others use lasers, jamming, or cyberattacks to disrupt a satellite without destroying it. Destructive ASAT tests are especially dangerous because the debris they create can threaten every satellite in that orbit, not just the one being targeted.

III. *Dual-Use Technology*

Many space technologies can be used for both peaceful and military purposes. For example, a satellite built to repair other satellites could, with small changes, be used to interfere with or disable them. This overlap makes it difficult to tell the difference between defensive and offensive intentions in space, which is one reason global rules on this issue remain limited.

Why These Issues Are Connected

Debris, uncontrolled satellites, and weaponization should not be seen as separate problems. More satellites lead to more congestion, which increases the risk of collisions and debris. As space becomes more important to security, countries may feel pressured to protect their satellites or develop stronger space capabilities, which can raise tensions further. Even actions meant to protect satellites can be seen by other countries as a security threat.

Why International Cooperation Is Needed

No single country can solve this issue alone. Space powers often keep their military activities secret, which makes cooperation difficult. Existing agreements, such as the Outer Space Treaty and the Liability Convention, provide some basic rules, but they were written decades ago and do not clearly address newer challenges like debris removal, satellite interference, and modern weapons technology. Countries also have differing abilities to track objects in orbit, which makes shared monitoring and trust difficult to build.

This agenda asks delegates to consider how the world can keep using space safely, fairly, and peacefully, while addressing the real risks posed by debris, uncontrolled satellites, and the growing military use of space. There is no single or easy solution, and progress will depend on cooperation between countries with different interests and capabilities. This is the responsibility your committee now holds.

IMPACTS DUE TO THE AGENDA

1. *Impacts of Orbital Debris Proliferation*

- **Collision risk and the Kessler Syndrome:** First theorised by NASA scientist Donald Kessler in 1978, the Kessler Syndrome describes a runaway scenario in which the density of objects in low Earth orbit (LEO) becomes so high that collisions generate debris at a rate faster than it can be removed, triggering a cascade of further collisions. Every fragmentation event increases the amount of trackable and untrackable debris. Some pieces are too small to be detected or tracked, but they can still be large enough to destroy a satellite if they hit it.
- **Threat to operational infrastructure:** Active satellites, including those providing GPS/GNSS navigation, weather forecasting, disaster response, and telecommunications, must increasingly perform collision-avoidance manoeuvres, consuming fuel and shortening operational lifespans. The International Space Station has conducted numerous debris-avoidance manoeuvres over its lifetime, and crews have periodically been required to shelter in return vehicles during close-approach events.
- **Economic cost:** Insurance premiums for satellite operators have risen in tandem with congestion in popular orbital shells. Debris remediation, tracking, and avoidance capabilities represent a growing and largely uncoded externality of space activity, disproportionately affecting new entrants and developing spacefaring nations who lack the tracking infrastructure of established powers.
- **Barrier to equitable access:** As orbital slots and debris-free paths become harder to find, countries and companies that entered space first gain an important advantage. This is similar to the “first come, first served” competition for resources on Earth. It raises fairness concerns for developing countries and nations that are only beginning their space programmes. This issue connects to UNSRA’s long-standing principle that space should benefit all countries, regardless of their level of economic or scientific development.

2. Impacts of Uncontrolled Satellites

- **Loss of custody without loss of liability:** Under Article VII of the Outer Space Treaty and the 1972 Liability Convention, the launching state remains internationally liable for damage caused by its space object even after the satellite has become defunct or uncontrollable. This creates a growing number of orbital “liabilities”—space objects that are still legally owned and someone is responsible for, but can no longer be controlled or tracked accurately.
- **Collision and re-entry unpredictability:** Uncontrolled satellites cannot perform avoidance manoeuvres, making them passive hazards to active missions and, upon natural decay of orbit, unpredictable re-entry risks to populated areas on Earth. Large derelict objects — spent upper stages in particular — have in past incidents re-entered the atmosphere with debris surviving to the ground.
- **Congestion of geostationary orbit:** Geostationary orbit (GEO) is a uniquely finite resource: a single ring above the equator where satellites can maintain a fixed position relative to Earth. Defunct GEO satellites that are not manoeuvred into designated "graveyard orbits" permanently reduce the number of usable slots. Failure to secure or responsibly retire uncontrolled satellites therefore compounds both the debris problem and the broader governance question of orbital slot allocation, historically managed by the International Telecommunication Union (ITU).
- **Attribution and forensic difficulty:** When an uncontrolled satellite breaks apart, collides with another object, or is suspected of being tampered with, establishing responsibility is complicated by incomplete registration data, ambiguous ownership after corporate mergers or state successions, and the technical difficulty of distinguishing accidental malfunction from deliberate interference.

3. Impacts of Space Infrastructure Weaponization

- **Dual-use ambiguity:** Much of the technology required to service, refuel, or de-orbit a satellite — rendezvous and proximity operations (RPO), robotic arms, grapple mechanisms — is identical to what would be required to disable or capture an adversary's satellite. This dual-use character makes verification and arms-control extremely difficult, and has already prompted several states to characterise routine servicing or debris-removal missions by rival powers as threatening.

- **Demonstrated ASAT capability:** Several states have conducted direct-ascent anti-satellite missile tests that destroyed defunct satellites in orbit, each time generating new debris clouds that persist for years and threaten all orbital users indiscriminately — including the testing state's own assets. These tests have often been criticised within the UN because they turn a short-term show of military power into a long-term danger for everyone using space.
- **Escalation risk:** Reliance on satellites for military communications, early-warning systems, and precision navigation means that an attack on space infrastructure could be interpreted as a prelude to, or component of, a wider conflict, raising the risk of rapid escalation between nuclear-armed states. The absence of a clear, binding definition of what constitutes a "weapon" or an "attack" in the orbital domain compounds this risk.
- **Non-kinetic threats:** Beyond kinetic ASAT weapons, states have developed or are suspected of developing electronic warfare, cyber intrusion capabilities against satellite command-and-control systems, and directed-energy or co-orbital "shadowing" systems capable of disabling a satellite without generating debris — but which are correspondingly harder to detect, attribute, and regulate

PAST TREATIES

The foundation of international space governance is a set of treaties developed under the United Nations. These treaties establish principles for responsible space activities, state responsibility, and the peaceful use of outer space.

The 1967 Outer Space Treaty (OST) is the cornerstone of international space law. It establishes that outer space is free for exploration and use by all States and prohibits the placement of nuclear weapons or other weapons of mass destruction in orbit. It also makes States responsible for national space activities, including those conducted by private entities. Its key principles include:

- Peaceful use: Outer space must be used for peaceful purposes and cannot become the territory of any State.
- State responsibility: Countries are responsible for their governmental and private space activities.
- Liability: States can be held responsible for damage caused by their space objects.
- International cooperation: States should share information and promote cooperation in space activities.
- Environmental protection: Space activities should avoid harmful contamination and interference.

Other Important Treaties and Frameworks:

1. **1972 – Liability Convention:** Makes launching States responsible for damage caused by their space objects, including satellites and rockets.
2. **1975 – Registration Convention:** Requires States to provide the UN with information about objects they launch into space, helping identify and track satellites and establish responsibility.
3. **2007 – Space Debris Mitigation Guidelines:** Provides recommendations for reducing the creation of orbital debris, including preventing accidental break-ups and safely disposing of spacecraft after their missions.
4. **2019 – Guidelines for the Long-Term Sustainability of Outer Space Activities:** Promotes safe and sustainable space operations through measures such as information sharing, collision avoidance, space-debris monitoring, and responsible satellite operations.

LEGAL ANALYSIS

Existing International Law — Charter and Jus ad Bellum. The UN Charter's Articles 2(4) and 51 govern any resort to force, and this applies to autonomous weapons as much as manned ones. AWS cannot lawfully be used to initiate aggression, and any self-defence use must still satisfy necessity and proportionality — the same jus ad bellum rules that already bind conventional weapons.

International Humanitarian Law (IHL). IHL is the area most directly implicated. The core principles of distinction, proportionality, necessity, and precautions must govern any use of force, and commanders need the real-time ability to make these legal judgements. If an AWS operates without adequate human supervision, it risks impeding that judgement and violating IHL.

The Martens Clause. Where no specific treaty rule exists, the Martens Clause fills the gap: civilians and combatants remain protected by “the principles of humanity and the dictates of public conscience.” The ICRC treats humanity's discomfort with removing humans from lethal decisions as a guide for interpreting the law, so even novel AWS must be judged against these broader humanitarian norms.

Weapon-Specific Prohibitions. IHL also bans certain weapons outright for being inherently indiscriminate or causing superfluous injury, such as blinding lasers. If an AWS cannot reliably distinguish combatants from civilians, it risks being treated as an unlawful weapon under this same category.

Human Rights Law. Outside active armed conflict — for instance in border control or policing — human rights law governs the use of force instead. Several states have argued at the UN that AWS without meaningful human control fundamentally challenge the right to life, and rights advocates point to further risks around privacy and non-discrimination.

International Criminal Law. International criminal law has no provisions specific to AWS yet, but existing war crimes and crimes-against-humanity law would still apply if an AWS malfunctions or is misused. The harder problem is assigning responsibility: it remains unclear whether commanders, operators, programmers, or manufacturers can be held liable for an autonomous system's unpredictable actions. This “accountability gap” is one of the strongest arguments for new binding rules.

Treaty Law and New Obligations. No treaty currently addresses LAWS specifically. States have two realistic paths forward: negotiating a new Protocol

under the CCW, or pursuing an independent convention. Either instrument would, under the Vienna Convention on the Law of Treaties, bind only its own parties. GA Resolutions 78/247 (2023) and 79/239 (2024) both show growing UN interest in pursuing one of these routes.

Jus Cogens Limits. Whatever model is chosen, any treaty must remain consistent with jus cogens — peremptory norms that cannot be reinterpreted or bargained away. The prohibition on indiscriminate force and the requirement to respect human dignity fall into this category, so no treaty on AWS, however innovative, can weaken existing IHL or humanitarian principles.

In sum, no law currently prohibits AWS outright, but the combined weight of IHL, human rights law, and criminal accountability rules makes it very difficult for fully autonomous, human-out-of-the-loop systems to meet existing legal requirements — which is precisely why so many actors are now pushing for new, binding norms.

Key Legal Issues

Distinction and Discrimination. IHL requires combatants to be reliably separated from civilians, but critics argue AWS cannot yet interpret context — intent, behaviour, or a wounded soldier's status — well enough to make that judgement safely. Some jurists counter that sophisticated sensors could eventually outperform humans here, but no consensus exists.

Proportionality. Attacks that would cause excessive civilian harm relative to military gain are prohibited, and this calculation traditionally depends on complex human judgement. Critics doubt an algorithm can weigh moral and emotional factors the way a person can, though others argue AI could apply quantitative proportionality rules more consistently than a poorly trained soldier.

Precautions and Targets. Parties must take all feasible steps before and during an attack to avoid harming civilians. An AWS operating without real-time supervision may fail to adapt to changing conditions or cancel a strike in time, making it essential to record and review every targeting decision for accountability.

Meaningful Human Control (MHC). Perhaps the most contested issue is what “meaningful human control” actually requires. Most proposals demand it, but the term has no fixed legal definition, and some states argue that focusing on “control” misses the point since autonomy itself will keep changing how warfare works. Debate continues over whether human-in-the-loop or human-on-the-loop oversight is sufficient.

Accountability and Liability. Existing law assumes a human is making the decision, but AWS create a real risk that no one is fully in control when something goes wrong. Many proposals would require states to designate a specific human operator responsible for each autonomous action, so crimes resulting from misuse can still be prosecuted.

Other IHL Weapons Rules. AWS must still meet the same weapon-specific rules that apply to any new weapon: they cannot be inherently indiscriminate or cause unnecessary suffering. Article 36 of Additional Protocol I requires every new weapon to undergo a legal review before deployment — likely a stricter review than conventional arms typically receive.

KEY STAKEHOLDERS

1. United States

The US leads in almost every way: most active satellites, best tracking systems, and the only country to formally give up one type of destructive weapons testing.

- In 2022, the US pledged to stop destructive anti-satellite (ASAT) missile tests, and has since gotten other states to back this at the UN.
- SpaceX's Starlink is the biggest satellite network in orbit; its military version, Starshield, shows how blurred the line is between commercial and defence space.
- US Space Command runs the world's best space-tracking system, and the US uses this advantage in its safety deals with other countries.

2. Russia

Russia has deep space heritage and, unlike the US, has actually tested a destructive ASAT weapon.

- In 2021, Russia blew up its own dead satellite (Kosmos-1408) with the Nudol system, creating 1,500+ pieces of debris and forcing the ISS crew to take shelter.
- Russia and China have jointly pushed the PPWT treaty at the UN since 2008, as a binding alternative to Western voluntary rules.
- Roscosmos is building closer ties with China on a joint lunar base, even as its role on the ISS fades.

3. China

China is racing ahead on two fronts: building huge satellite networks to rival Starlink, and holding the record for the worst debris event ever.

- China's 2007 ASAT test destroyed its own dead weather satellite, creating over 3,000 tracked debris pieces — still the largest debris event on record.
- Its Guowang and Qianfan satellite networks plan for tens of thousands of satellites; several hundred were already up by mid-2026.
- Western militaries believe China's Strategic Support Force runs its own counter-space systems alongside the civilian programme.

4. India

India sits in the middle: it has tested an ASAT weapon, but also runs one of the more transparent debris-mitigation programmes.

- In 2019, India's Mission Shakti destroyed its own satellite at a low altitude (~300 km), a choice ISRO defends as limiting how long the debris stays in orbit.
- ISRO's IS4OM system, running since 2022, tracks India's satellites and manages collision-avoidance.
- India joined the Artemis Accords in 2023, while still keeping its long-standing space ties with Russia.

Other Stakeholders

1. EU / European Space Agency (ESA)

The EU and ESA lead the main push for binding debris-reduction rules. In 2024, 12 states signed ESA's Zero Debris Charter, aiming for debris-neutral operations by 2030; by 2026 that had grown to 21 states and 220+ organisations. The EU also wants an EU Space Act to regulate satellite operators. But ESA has no military of its own, and its members don't always agree on security matters, which limits how united Europe can be on weaponization.

2. Private Space Industry

Companies now shape orbit as much as governments, even though treaties technically bind states, not companies. Starlink is the largest network ever built, alongside China's Guowang and Qianfan and Amazon's Kuiper — all launching satellites faster than governments can regulate them. These companies set the real rules on debris and collision-avoidance through their own policies, even though they can't sign a resolution themselves.

3. United Nations (COPUOS / UNOOSA)

As the parent body of this committee, COPUOS produced the two main guideline documents in this area: the 2007 Debris Mitigation Guidelines and the 2019 Long-Term Sustainability Guidelines. UNOOSA supports the Committee and keeps the official registry of space objects. Neither can enforce anything, and since COPUOS works by consensus, one objecting state — as has happened for years with the PPWT — can block progress indefinitely.

4. Developing and Non-Spacefaring Nations

Countries without their own launch capability still have a stake here — mainly in how orbital slots and radio frequencies are shared out through the ITU. Many worry that mega-constellations from a few big powers will grab the best orbits before smaller states get the chance to use them, going against the Outer Space Treaty's promise that space belongs to 'all mankind.' These states usually push COPUOS for more support and fairer access, rather than new limits on the major powers.

Comparative Actor Profiles

KEY ACTOR PROFILE	UNITED STATES	RUSSIA	CHINA	INDIA
Positioning	Leader on 'responsible behaviour'; leads Artemis Accords; avoids binding treaties.	Backs binding PPWT; defends 2021 test as sovereign right; leaning on China.	Calls build-out civilian; West sees military use too; backs PPWT.	Responsible, rising power; balances US and Russia ties.
Capacity	Space Force runs top tracking network; classified counter-space capabilities.	Nudol ASAT + 'inspector' satellites; under-resourced vs US/China.	Strategic Support Force counter-space systems; fast-scaling launches.	ASAT capability (2019); runs IS4OM and NETRA tracking.
Diplomatic Tools	UNSC veto; Artemis Accords; ITAR export controls; data-sharing deals.	UNSC veto; PPWT sponsor; lunar/security ties with China.	UNSC veto; lunar partnership with Russia; Belt and Road diplomacy.	Artemis Accords (2023); active IADC member; growing launch business.
Legal Stance	Cites OST Article VI; no Moon Agreement; claims resource-extraction right.	Core UN treaties, not Zero Debris Charter; rejects 2021 test criticism.	No Zero Debris Charter/Artemis; 'province of all mankind' argument.	No Moon Agreement; favours COPUOS guidelines over PPWT.

QUESTIONS A RESOLUTION MUST

ANSWER [QARMAs]

1. How can UNSRA establish effective authorization and supervision for private entities operating debris-generating or weaponisation missions, in line with Article VI of the Outer Space Treaty?
2. How should the committee distinguish lawful dual-use space activity (servicing, debris removal, proximity operations) from hostile military action?
3. Who is financially and legally responsible for removing existing debris and defunct satellites already in orbit?
4. Should destructive ASAT weapons testing be banned or restricted by treaty, and how would compliance to the action taken be verified?
5. How can access to space situational awareness (tracking) data be made more equitable for non-spacefaring and developing states?
6. What obligations should private satellite operators bear to avoid uncontrolled threats, given that states remain internationally liable for their conduct?
7. Does Article IV of the Outer Space Treaty need to be expanded to explicitly cover conventional and non-kinetic weapons in orbit?
8. How should national space agencies coordinate re-entry notification for uncontrolled objects to satisfy the "international cooperation" principle of the Outer Space Treaty?

CASE STUDIES

1. Iridium 33–Cosmos 2251 Collision (2009)

On 10 February 2009, a major collision occurred between Iridium 33, an active American commercial communications satellite, and Cosmos 2251, an inactive Russian communications satellite, approximately 790 km above Earth. The two satellites collided at a relative speed of around 11.7 km per second, completely destroying both spacecraft and generating thousands of pieces of orbital debris.

The incident was particularly significant because Cosmos 2251 was no longer operational or actively controlled. It demonstrated that satellites can remain a serious threat even after their missions have ended. The debris produced by the collision continued to travel through orbit at extremely high speeds, increasing the risk of further collisions with operational satellites and spacecraft.

The incident highlighted the importance of accurate satellite tracking, collision-avoidance systems, international sharing of orbital information, and responsible end-of-life disposal of satellites. It also raised questions about responsibility when an inactive or uncontrolled satellite causes damage to another country's space infrastructure.

2. Fengyun-1C Anti-Satellite Test (2007)

On 11 January 2007, China conducted an anti-satellite (ASAT) test in which a missile was used to destroy its Fengyun-1C weather satellite at an altitude of approximately 865 km. The destruction generated thousands of pieces of trackable orbital debris, along with numerous smaller fragments that were difficult to detect and monitor.

The incident became one of the largest debris-generating events in the history of space activities. Since the debris remained in orbit, it posed risks to satellites and spacecraft operated by countries that were not involved in the original test. This demonstrated how a military action carried out by one country can create long-term consequences for the entire international space community.

The test also increased international concerns regarding the weaponization and militarization of outer space. Although the Outer Space Treaty prohibits weapons of mass destruction in orbit, it does not comprehensively prohibit conventional ASAT weapons, leaving an important area for further international discussion and regulation.

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